

Emulation or differentiation? China's development finance and traditional donor aid in developing countries*

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Abstract

Foreign aid relationships are valuable to donors as a means of improving development outcomes and influencing recipient country policy. The emergence of new donors can lead to competition as donors vie for recipient governments' attention. How does such competition affect the behavior of traditional donors toward developing countries? I draw attention to how the rise of China as a provider of development finance is changing the *type* of development that traditional donors support. Chinese development finance is particularly targeted at large infrastructure projects, and this focus can exert pressure on traditional donors. I suggest traditional donors can either emulate China's approach to development, i.e. offer large, infrastructure-intensive projects, or differentiate themselves and specialize in alternative approaches to development, e.g. focus on governance and social sector interventions. I test this using data on the terms of World Bank, US, and Chinese development finance in over 100 countries. I find the World Bank responds to competitive pressure from China by emulating the Chinese emphasis on infrastructure, allocating a greater share of its development projects in infrastructure-intensive sectors in the year after recipient countries receive more Chinese development finance. Meanwhile, the US is less likely to respond to China's increased presence. China's growing role as a provider of development finance is affecting traditional donor behavior, shaping the type of development donors support by introducing bottom-up pressure at the recipient country level.

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1 Introduction

Since the end of the Cold War, advanced economies have been largely alone in providing development finance to developing countries, whether bilaterally or through multilateral institutions. For donors, these relationships with recipient countries offer a means of improving development outcomes as well as influencing recipient country policies. However, the rise of emerging donors, especially China, has disrupted the virtual monopoly previously enjoyed by traditional donors.¹ Given that recipient governments have a finite capacity to absorb external finance and limitations on the time they can allocate to different donors and creditors, alternative sources of development finance can place competitive pressure on traditional donors. In this paper I investigate how traditional donors respond to the arrival of China as a donor, contributing to a growing literature on competition among donors as the landscape of development finance becomes more crowded (Granath 2016; Hernandez 2017; Swedlund 2017a). I focus on the competitive pressure arising from the focus of Chinese development finance on large infrastructure-intensive projects and suggest traditional donors can respond to China’s growing role by either emulating China’s approach to development finance or differentiating the development assistance they offer. Is competition with China changing the type of development that traditional donors pay for?

In 2015, the World Bank announced it would provide a \$700 million guarantee to assist the construction of the Sankofa natural gas plant in Ghana. The guarantee was the largest ever of its kind from the World Bank anywhere in the world, and the Bank heralded it as a “transformational project” for Ghana’s energy sector (World Bank 2015). China’s growing role in Ghana was likely a factor in the Bank’s decision to provide this unprecedented guarantee. Only a few years earlier, the Ghanaian government agreed a large Chinese-financed gas project, the Atuabo gas processing plant. The Government of Ghana had initially also planned to develop this earlier gas project with the World Bank. However, a few years into preparations with the Bank, and after elections brought in a new government, Ghana decided instead to finance the project with the China Development Bank, securing a \$1 billion loan to develop the facility with a Chinese contractor (Boadu and Aklorbortu 2015). Speculation in the press and in policy circles was that the Ghanaian government considered the World Bank’s environmental safeguard requirements too onerous, and instead chose Chinese financing for its speed and ease. That the World Bank provided an unprecedented guarantee for a similar project just a few years later is an example of how competitive pressure from China may lead traditional donors to *emulate* China, taking on projects that are similar in size and scope to the large infrastructure projects supported by China.

However, emulation of China’s approach to development projects is not the only possible response to competitive pressure from Chinese financing. Instead, donors may choose to *differentiate* their approach. For example, when Ethiopia and the EU negotiated the focus of

¹ The terms “donor”, “aid” and “development finance” have become murkier as more countries have become involved in the international practice of providing finance for development. In particular, China eschews the label of “donor,” since it prefers to emphasize the “mutual benefit” of its development finance. For ease of exposition, I refer to the World Bank and other long-standing providers of development finance as “traditional donors” throughout, to contrast them with emerging actors, especially China. I refer to “development finance,” since not all official finance, especially from emerging donors, meets the conventional definition of “aid”, which requires that it be highly concessional or in the form of grants.

development cooperation for 2014-2020, the EU announced its intention to withdraw from the road sector, where it had been active as a donor since the 1990s. Though noting that “the road sector [in Ethiopia] remains one of least developed in Sub-Saharan African countries,” the final cooperation agreement states that the EU intends to shift its financing out of the roads sector and into other sectors, especially social sectors (Government of Ethiopia and European Commission 2014: 14). In the new plan, over 60% of the EU’s resources will be dedicated to sustainable agriculture and health, and resources previously dedicated to transport are to be shifted into renewable energy and governance reforms in the energy sector. This shift occurred against the backdrop of a growing Chinese presence in the Ethiopian road sector, including a \$320 million loan from the China Export Import Bank to finance a ring road around the capital Addis Ababa in 2013.² When confronted with competition from China, traditional donors may also choose to *specialize* in alternative sectors rather than mimicking the types of projects financed by China.

China’s infrastructure-led approach to its own development has received considerable attention, as have its efforts to export this approach abroad by providing funding for similar infrastructure investments, including through the One Belt One Road initiative (Ansar et al. 2016; Fukuyama 2016). I investigate how these traits of Chinese development finance affect the behavior of traditional donors, suggesting that China may have an effect on global norms and practices in development through the competition it introduces at the recipient country level, spurring traditional donors to compete with China and shift their funding to different kinds of projects. Previous work on the effects on donor behavior of a growing diversity of donors has focused on whether donors reduce the conditionality of their aid in response to Chinese competition, since China differs notably from traditional donors in the lack of policy and governance conditions attached to its development operations (Hernandez 2017).

Building on this existing work on aid conditionality, I instead draw attention to other distinctive attributes of Chinese financing that can exert competitive pressure on traditional donors. These are the large size of Chinese development projects and their focus on infrastructure, especially in transport and energy sectors. Recipient governments have a preference for such projects, which help to meet infrastructure funding gaps and are often politically useful for incumbents as highly visible examples of development achievements (Swedlund 2017a: 404-405). China’s approach to financing development thus places traditional donors under competitive pressure by offering alternative development financing that is attractive to recipients. Since recipients have finite time to dedicate to meeting with donors, and finite capacity to take on external finance, China’s growing role has the potential to reduce the relevance of traditional donors.³

I argue that three things combine to place competitive pressure on traditional donors: the differences in the policy focus of Chinese development finance and traditional donor aid, recipients’ preferences for “Chinese-style” development finance, and the zero-sum nature of competition for access to recipient governments. Furthermore, I contend that traditional donors can respond to this competition from China’s development orientation in two ways: emulation or differentiation. Traditional donors could imitate the Chinese approach to development finance, shifting their development projects into infrastructure-intensive productive sectors and offering larger

² By comparison, the EU had previously allocated roughly EUR 280 million to the *entire* road sector in Ethiopia over a period of 6 years, 2008-2013.

³ Though the constraints in taking on debt might be most obvious, there is also a limit in the amount of concessional finance or grants a country can take on. This is discussed further in Section 3 below.

projects. Alternatively, they could differentiate themselves from the Chinese approach, focusing on social sector projects in health and education that are complementary to Chinese finance. There are incentives for both possible responses: emulation may be the most rapid means of responding to the competitive threat of China, but differentiation may make use of traditional donors' comparative advantage in capacity building and projects in social sectors.

I study these dynamics with respect to both the World Bank, a large multilateral donor, and the United States, a significant bilateral donor likely to perceive competition with China in the context of larger geopolitical competition. In the case of the World Bank, competition with China plausibly arises out of reduced access to recipient governments and a pressure to extend loans to meet lending targets, while for the United States, competition with China is more likely to be seen in a global strategic context. Even for the World Bank, however, strategic considerations of donor countries may enter the decision-making process, as research has shown the extent to which influential shareholders can affect the behavior of international financial institutions (Kilby 2009; Stone 2004, 2008; Vreeland and Dreher 2014). I use two measures to ascertain whether these traditional donors respond to China using emulation or differentiation: the number of large projects traditional donors extend to recipient countries (greater than \$50 and \$100 million) and the share of traditional donors' development projects in infrastructure-intensive sectors, including energy, transport, and industry.

I use data on the terms of World Bank and American development projects in over 100 countries over the period 2000-2014. My key explanatory variable is Chinese official financing commitments, taken from the AidData Global Chinese Official Finance dataset (Dreher et al. 2017a). To address concerns about the endogeneity of Chinese finance, I also present results using an instrumental variable approach that uses Chinese steel production as an instrument for Chinese development finance, which has been previously successfully applied by Dreher et al. (2017a) in their work on the effects of Chinese finance on economic growth. Throughout the paper, I include illustrative anecdotes taken from over 140 interviews carried out in Ghana (August 2013, April-May 2017), Kenya (January-February 2017) and Ethiopia (July-August 2017).

I find evidence that the World Bank is responding by emulating the Chinese approach to development finance, especially in the sectoral allocation of their aid. When recipient countries receive more Chinese development finance, the World Bank allocates more of its development projects to infrastructure-intensive sectors, such as transport or energy. By contrast, the US seems less likely to emulate the Chinese approach to development in response to competition, though there is some evidence that it extends a greater number of large projects to countries that receive Chinese financing.

The paper proceeds as follows: Section 2 briefly discusses the rise of Chinese development finance and compares key attributes of Chinese and traditional donor financing. Section 3 argues that competition from China can directly affect the types of projects traditional donors finance in developing countries, because traditional donors can choose to respond to competition either by emulating China and converging on large projects targeted at productive sectors or to respond by specializing in alternative and potentially complementary development approaches. Section 4 discusses the data and the methods employed, Section 5 presents and analyses the results, and Section 6 offers a brief conclusion.

2 Introducing competitive pressure? Characteristics of Chinese finance

2.1 Characteristics of Chinese development finance identified in the existing literature

China has substantially increased its official financing to developing countries over the past decade, thereby raising the likelihood of competition with traditional donors. Data from AidData’s Global Chinese Official Finance dataset shows the steady growth of Chinese overseas financing commitments from a low base in the early 2000s. China committed \$2.6 billion in official finance in 2000, an annual commitment that expanded to over \$36 billion by 2013.⁴ Over the past decade, Chinese official financing commitments have been catching up to the World Bank and the US’s global portfolio of development projects, even surpassing the Bank in some of the recent years. In some regions, especially in sub-Saharan Africa, Chinese finance is exceeding that of the World Bank, and even the US in some years (Figure 1).

⁴ In both the descriptive statistics that follow and the analysis reported in section 5, I use data on donors’ financing commitments, rather than disbursements. In part, this is due to data availability. The AidData Global Chinese Official Finance dataset uses the “Tracking Underreported Financial Flows” (TUFF) methodology, which provides data only on commitments, not disbursements of Chinese finance. For direct comparison, I therefore use data on World Bank commitments, rather than disbursements, though these can differ. However, given my interest in competition, commitments are an appropriate measure. While studies of the *effects* of aid and finance require data on the amount of finance actually received, commitments data is suitable for studying the *intentions* of donors. Disbursements can sometimes vary for reasons that are out of the control of donors, while commitments are fully determined by donors.

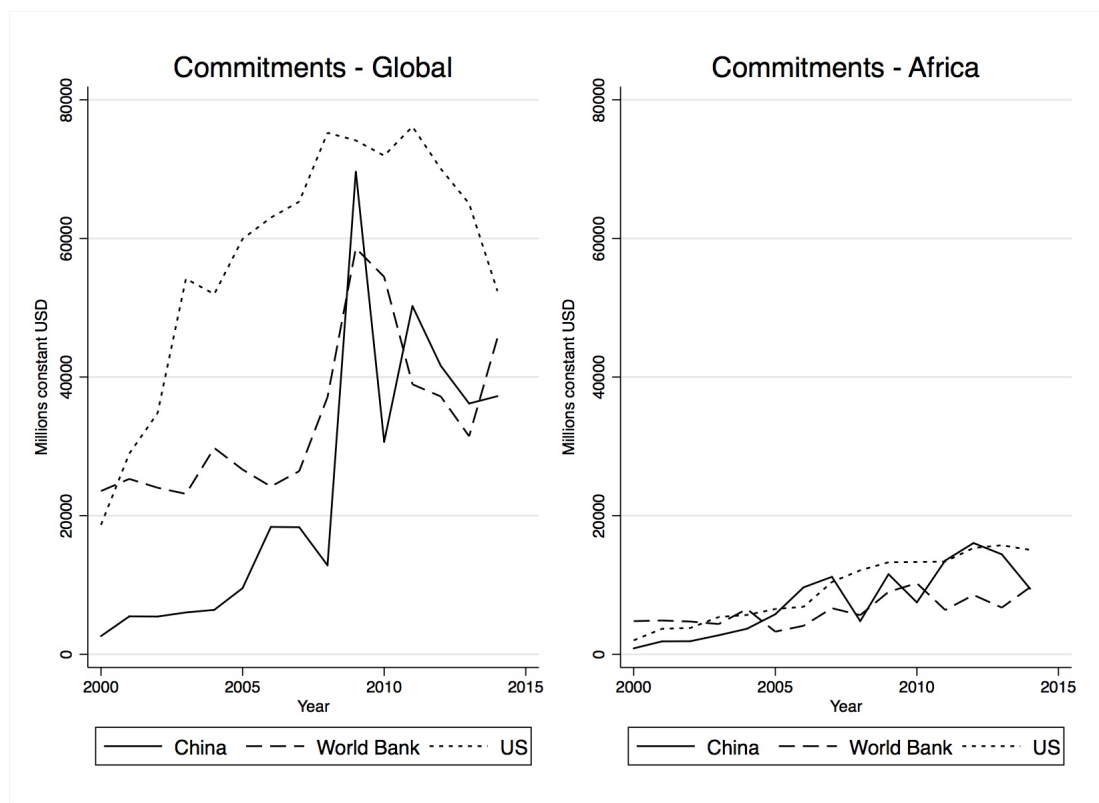


Figure 1: Official overseas finance commitments and number of development projects: China, the World Bank, and the US, 2000-2014.

Sources: AidData Global Chinese Official Finance Dataset Version 1.0 (2017); World Bank Projects & Operations Dataset (2017); USAID, Foreign Aid Explorer: The official record of US Foreign Aid (2017).

Chinese official financing has a wide regional scope, though Asia and Africa have received the bulk of grants and credits (Figure 2). Over the period 2000-2014, African countries received a cumulative \$118 billion in official financing from Chinese sources, while the total in Asian countries amounted to \$116 billion. In Latin America and the Caribbean, financing commitments over the period totaled \$56 billion, and in Central and Eastern Europe the total came to \$53 billion.

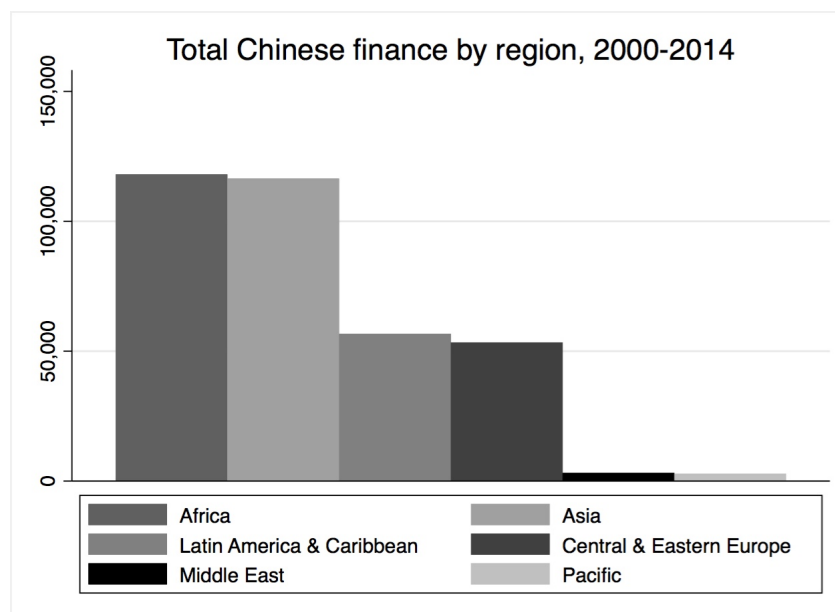


Figure 2: Regional breakdown of total Chinese official finance commitments, 2000-2014.

Source: AidData Global Chinese Official Finance Dataset Version 1.0 (2017).

Note: Legend should be read left to right.

A considerable body of scholarship has emerged to explain the expansion of Chinese development finance. Most trace the growth in Chinese overseas financing to China's growing global ambitions, an accumulation of dollar reserves arising from China's significant trade surplus and a desire to support Chinese firms expanding abroad. Critics often assert that Chinese aid is more commercially and politically motivated than that of traditional donors, particularly that it is targeted towards countries rich in natural resources, politically aligned with China or governed by autocratic rulers (Naím 2007; Rotberg 2008). However, Dreher and Fuchs (2015) find no effect of natural resource wealth or democracy on the allocation of Chinese concessional funds that most closely resemble traditional donors' aid. Similarly, Dreher et al. (2017b) stress that Chinese concessional commitments are not motivated by natural resource wealth or decisively shaped by regime type and institutional quality, though less concessional forms of official finance do appear to be driven by commercial considerations, including natural resource wealth.

One notable difference between Chinese development finance and traditional donor aid is the lack of policy conditionality attached to Chinese projects. While traditional donors often attach reform or policy requirements to their development assistance, frequently as a precondition for releasing aid to the recipient government in question, China has emphasized "non-interference" and mutual respect for sovereignty as central to its development program, and has openly distanced itself from the conditionality approach employed by traditional donors (Alden 2007; China 2011; Six 2009). Though Chinese official finance often comes in the form of export credits from China's ExIm bank, and thus requires the project in question be carried out by a Chinese firm, there are no policy or reform conditions beyond these commercial requirements. Hernandez (2017) has drawn attention to how this difference between Chinese development finance and traditional donor aid might place pressure on traditional donors, leading them to reduce the conditionality of their aid. He finds that the World Bank attaches fewer conditions to its loans when a recipient country receives greater commitments of Chinese finance.

2.2 Other distinctive characteristics of Chinese finance: Size and sector focus

There are additional distinctive attributes of Chinese development finance beyond the lack of policy conditionality, which I argue also have the potential to disrupt traditional donor behavior. Namely, Chinese development projects tend, on average, to be large and targeted towards infrastructure projects, especially in energy and transport sectors. Though these differences between Chinese development finance and traditional donor aid have been recognized in the literature, their effect on traditional donors has yet to be empirically investigated. By drawing attention to these differences in the sectoral allocation and modality of different donors' development finance, I am heeding the advice of Nielson et al. (2017), who encourage researchers to further disaggregate analyses of development finance by sector and modality.

These differences between Chinese development finance and traditional donor aid are attributable to a number of factors. First, since export credits make up a large part of Chinese official finance, there is a tendency towards projects drawing on Chinese companies' competitiveness in construction (Brautigam 2011). Second, since a considerable portion of Chinese financing is non-concessional, projects must have the potential to earn returns that will allow repayment. Therefore, there is a bias towards projects with a clear economic return, leading to a greater emphasis on infrastructure and productive sectors over health or education.

Furthermore, China stresses that its development finance is "demand-led" and responds to recipient countries' particular appetite for infrastructure projects. This combines with China's emphasis on infrastructure-led growth in its own development. The Chinese discourse on its development assistance stresses the potential for infrastructure projects to have a transformative effect on the economy (Kragelund 2014). This differs from the approach of many of the traditional donors, whose portfolio tended towards social sectors and smaller projects over the course of the 2000s, as aid budget shrank and the Millennium Development Goals emphasized achieving social sector objectives (Swedlund 2017b: 9). While this trend was less pronounced for the World Bank and other multilateral development banks, they too shifted funds into social sector programs in education and health.⁵ In particular, the increased sensitivity to environmental and social impacts of large-scale infrastructure projects has meant that funding such projects with the World Bank now requires extensive and lengthy environmental and social impact reviews (Park 2007). By contrast, Chinese funders are reputed to move much more quickly, without formalized and time-consuming environmental review procedures.

⁵ In fact, the World Bank's perceived neglect of infrastructure projects and productive sectors has been a long-standing concern for some recipient countries, as Kellerman (2018) describes with respect to Asian countries' criticisms of the World Bank in the 1960s.

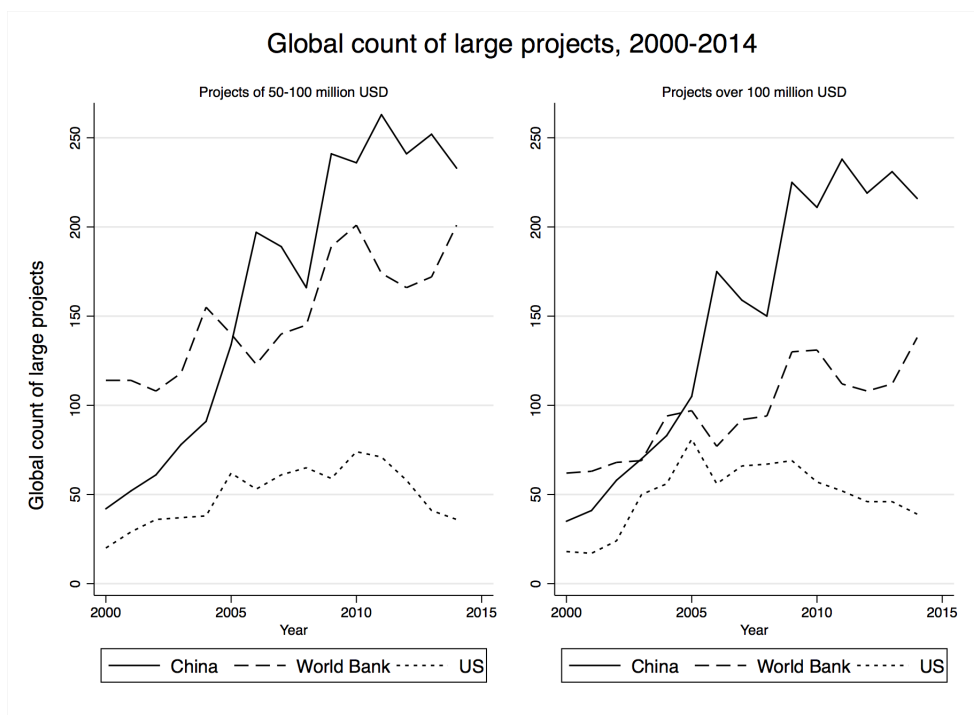


Figure 3: Count of projects of 50-100 million USD and over 100 million USD, 2000-2014

Sources: AidData Chinese Global Official Finance Database Version 1.0 (2017); World Bank Projects & Operations Dataset (2017); USAID, Foreign Aid Explorer: The official record of US Foreign Aid (2017).

These differences between Chinese and traditional donors' approaches to development projects are reflected in the data. Figure 3 shows the number of large projects supported annually by China, the World Bank, and the US. Over the period 2000-2014, China financed an average of 165 projects per year of \$50-100 million each, while the World Bank and the US supported only 151 and 49 projects of this size, respectively. The gap becomes even clearer when looking at very large projects of over \$100 million. China financed an average 148 projects per year of this size, while the World Bank and the US undertook only 96 or 50 projects this large, respectively. The Chinese emphasis on “megaprojects” thus distinguishes its development finance from that of traditional donors, potentially introducing competition.

Chinese projects are also distinctive in their sector focus. Figure 4 shows the proportion of annual financing donors allocated to projects in “hard sectors” such as energy, transport or industry, where infrastructure projects are more likely.⁶ Over the period 2000-2014, an average of 20% of Chinese projects per year were allocated to infrastructure-intensive “hard sectors,” while only 11% of yearly World Bank projects and 8% of annual US development projects were allocated to similar sectors during that time. Data for Chinese development finance and for

⁶ The World Bank's sector classification is different from that used in the AidData Chinese Official Finance dataset and the USAID dataset, but effort has been taken to make these as comparable as possible. The World Bank's internal sector classification structure comprises eleven sectors, while the AidData China dataset and the USAID dataset use the 24 sector categories of the OECD's Creditor Reporting System. Furthermore, World Bank projects may be associated with multiple sectors, while both the Chinese and USAID projects are only assigned to a single sector. I coded World Bank projects as being in “hard sectors” if their only “major sector” was one of the following: (1) energy and extractives, (2) industry, trade and services, and (3) transport. I coded Chinese and American projects as in “hard sectors” if they are classified as (1) water and sanitation, (2) transport, (3) communications, (4) energy, (5) agriculture, (6) industry, mining and construction.

sectoral allocation of US aid is only available from the year 2000, but for the World Bank it is available going back to 1980. The historical data on the World Bank projects shows the decline in World Bank projects in “hard sectors” over the course of the late 1980s and 1990s, and the subsequent increase in the late 2000s. This most recent increase in World Bank projects in “hard sectors” may be attributable to emulation of Chinese development practices in response to competition, but may also reflect an intellectual shift within the World Bank over the last decade, as it has increasingly come to emphasize economic transformation as a driver of development.

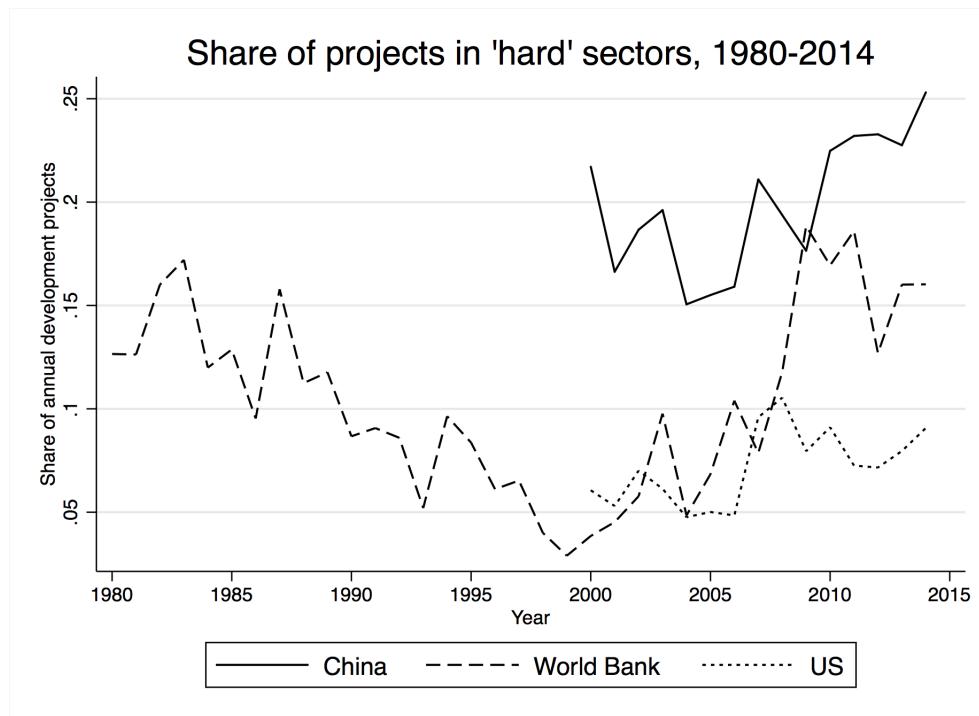


Figure 4: Share of US, World Bank, and Chinese development projects in “hard” sectors, i.e. energy, transport or industry. Data for the World Bank is available 1980-2014, for the US and China only for 2000-2014.

Sources: AidData Global Chinese Official Finance Dataset Version 1.0 (2017); World Bank Projects & Operations Dataset (2017); USAID, Foreign Aid Explorer: The official record of US Foreign Aid (2017).

3 Emulation or differentiation: Traditional donor responses to Chinese finance

The entrance of China as a new donor with different practices to those of traditional donors can introduce competition and disrupt the relationship between recipient governments and traditional donors. I suggest the entry of China has the potential to re-shape the type of development that traditional donors choose to support with their funding. Donors can respond to competitive pressure by either emulating the Chinese approach to development finance and funding large projects in infrastructure-intensive sectors, or differentiating themselves by specializing in alternative niches.

3.1 Competition for recipients

The claim that the entrance of a new donor can lead to competition implies donors are contending for access to a scarce resource. Following others, I suggest donors seek access to the recipient government in order to shape government policy in recipient countries. Since access to recipient countries is (mostly) zero-sum, this introduces competition.

There is a considerable literature on donors' motivations in giving aid (Alesina and Dollar 2000; Bermeo 2011, 2017; Dreher et al. 2013; Fuchs et al. 2014; Hoeffler and Outram 2011). One motivation identified in this literature is donors' interest in influencing recipient policy (Bueno de Mesquita and A. Smith 2016). Donors may value their influence over recipient policy for self-interested reasons, such as securing recipient governments' support in international organizations or favorable conditions for investors, or they may value their influence because they believe their research and experience across developing countries allow them to be effective in achieving development outcomes (Swedlund 2017b: 27). The latter is more relevant to multilateral donors, such as the World Bank, who seek access to recipients to translate their expertise into maximum development impact (Milner 2006: 109).⁷

To have such influence, donors need access to the recipient government, access they receive in connection with the development projects and programs they fund. In donor circles, this is referred to as "policy dialogue". Donors value their access to recipient governments and worry when exposure to senior policymakers in recipient countries is reduced. For example, in an interview, a donor representative in Kenya referred to access to government officials, in this case the President, as an indication of the overall relationship: "All donors feel the quality of the relationship is overall not as good as it should be... but recently there is more attentiveness in the Kenyan government to the importance of donors. For instance, our minister was able to meet with President Kenyatta over the weekend during his visit".⁸

Competition arises when the recipient government allocates its time to meeting with new donors, thereby reducing the access other donors have to encourage the government to pursue their desired policies. Traditional donors have sought to mitigate the challenge of zero-sum access to the recipient government by coordinating their positions, such that one donor's access to the government does not necessarily take away from others' access.⁹

In the case of China, however, access to recipient governments is essentially zero-sum, since China does not participate in coordination and harmonization mechanisms with other donors (Bigsten and Tengstam 2015). While a recipient government's time spent with an OECD donor or the African Development Bank likely reinforces the policy messages of donors such as the US or the World Bank, this is not the case with China. When recipient governments choose to dedicate time to meeting with China because they consider Chinese financing attractive, this reduces the access the World Bank or the US have to engage the government on their desired

⁷ Note that while much of the literature on multilateral and bilateral aid expects multilateral aid to be less politicized and more transparent than bilateral aid, Findley et al. (2017) find that recipients do not react differently to bilateral and multilateral aid.

⁸ Interview, Donor representative. Nairobi, Kenya, February 2, 2017.

⁹ In the 2005 Paris Declaration, donors in the Organization for Economic Cooperation and Development (OECD) Development Assistance Committee (DAC) committed to "harmonizing" their aid programs. However, coordination has in practice not been as extensive as promised. See Aldasoro et al. (2009) and Nunnenkamp et al. (2013).

policy outcomes. In Ghana, for example, one donor representative explained that the Ghanaian government was slow to respond to donor requests because they were occupied with Chinese loan negotiations: "One document was missing to allow [us] to disburse, and in spite of several requests to sign these papers the government didn't feel it was important because... they were negotiating the Chinese loan".¹⁰

A similar drive to compete is modeled by Bueno de Mesquita and A. Smith (2016), who focus explicitly on the competition arising from aid-for-policy deals. They assume donors provide aid in exchange for the recipient adopting policies desired by the donor. Competition arises when a rival donor offers aid in exchange for the recipient adopting a different policy. Since the recipient can only adopt one of these two policies, the outcome is necessarily zero-sum, and competition occurs. They find that donors are driven to provide increasing amounts of aid to secure their desired policy outcome. Using data spanning the periods before, during and after the Cold War period, they find that US aid allocation increases during periods of global competition with the Soviet Union.

Such competition among bilateral donors was very overt in the Cold War context, with the Soviet Union and the US deploying aid to win over developing country allies in a zero-sum game shaped by the broader geopolitics of international competition (Bermeo 2015). While aid competition between the United States and China is less heated, and aid relationships less zero-sum, closer relationships between developing countries and China do take on a geopolitical dimension for the United States. As Secretaries of State, both Hillary Clinton and Rex Tillerson warned African leaders about the growing role of China, with Clinton alluding to Chinese influence on the continent as "new colonialism," and Tillerson advising leaders to "carefully consider the terms of [Chinese] investments" (Martin 2018; D. Smith 2012). Given the geopolitical tenor of competition between China and the US, the US has particular incentives to respond to China's growing influence by making its development aid attractive to recipient countries.

These assumptions of zero-sum competition can be relaxed somewhat with respect to the World Bank's response to China. It is not always the case that China and multilateral donors such as the World Bank are pursuing diverging objectives in recipient countries. In fact, in a number of cases China ExIm Bank and China Development Bank have co-financed programs with multilateral banks such the African Development Bank. However, since China mostly does not coordinate its projects with the World Bank, the more time that recipients choose to dedicate to Chinese development projects, the fewer opportunities there are for the World Bank to pursue its own policy agenda with the recipient government. Thus, in order to retain their access to the government, the World Bank is incentivized to compete either by emulating the approach of Chinese development projects or by specializing to develop a complementary niche that attracts recipients.

In addition to the motivation to retain access to the government for "policy dialogue," there are bureaucratic drivers of competition, which may be more pronounced for the World Bank than for the US. Donors compete for a share of the total development finance a country receives. As other work has highlighted, World Bank staff are under pressure to disburse funds.¹¹ Yanguas and Hulme (2015) cite one World Bank practitioner who refers to the "overwhelming

¹⁰ Interview, Donor representative. Accra, Ghana, October 7, 2013.

¹¹ This is often discussed as an impediment to conditionality, since the Bank cannot credibly threaten to withhold funds in the case of non-compliance with conditions. See Svensson (2003)

pressure to get money out of the door” and that “task teams are paid to ensure that the money is disbursed”. Given this incentive to disburse, competition arises when recipients choose to accept development finance from other sources and then, for reasons of debt carrying capacity, reduce the amount they take from the World Bank.

This is most obvious with development finance that comes in the form of loans, such as those offered by the China ExIm Bank or the World Bank’s International Bank for Reconstruction and Development (IBRD). Recipient governments are constrained in the amount of debt financing they can take on by the need to keep their total debt below a sustainable threshold.¹² With loan finance, therefore, competition arises because recipients can only borrow so much, and the choice of one lender therefore reduces the amount that another lender can lend. Combined with bureaucratic incentives to disburse, this can lead to competition among providers of development finance for the share of total external finance that a recipient takes on.

It may appear that such competition is less likely with grant aid and concessional finance, since recipients are not constrained by debt sustainability concerns in their decision of how much aid to receive. However, there are also limits to the total amount of grant aid a recipient can take on. These limits arise not from debt sustainability, but rather from the transaction costs associated with development projects and the absorptive capacity of the recipient’s economy. Aid projects pose significant administrative costs for the recipient government, including the design of the project, monitoring of funds and reporting to donors (Paul and Vandeninden 2012). In some cases, recipients are asked to provide a financial contribution, referred to as counterpart funding, towards the project funded with donor grants. Limits on the capacity and resources of the recipient will thus act as a constraint on the total amount of grant aid a country can take on. These limits combine with the bureaucratic incentives to disburse and the desire for policy access to produce the potential for competition.

In sum, competition arises from the fact that donors are seeking access to the recipient government, access that is constrained by limits on the recipient government’s time and its ability to take on outside finance. This means I expect to see an effect of increasing Chinese finance on traditional donor behavior even though Chinese development finance and traditional donor aid tend to be quite different. In fact, it is precisely *because* Chinese development finance is different from traditional donor aid that I expect increasing Chinese finance to place competitive pressure on traditional donors to change their own development finance.

In her work on the effect of Chinese development finance on traditional donors’ aid negotiations, Swedlund (2017a) suggests that the rise of Chinese finance does not affect recipient governments’ bargaining power because Chinese development finance and traditional donor aid are different and not substitutable, and therefore not in direct competition with each other. However, since donors are competing for access to the recipient government and they gain access by providing development finance that is attractive to recipient governments, differences in the *type* of development projects may be the very reason that donors experience a competitive pressure to change the focus of their development finance. Whether a donors’ different development focus introduces competitive pressure depends on whether the recipient government finds this focus attractive, which is why I next turn to recipient preferences.

¹² Bunte (2013) makes a similar point about the impact of debt constraints, highlighting that these constraints mean that a choice for one lender is a choice against another lender.

3.2 Recipient preferences

Having established donor motivations that can lead to competition, it is important to specify recipient preferences with respect to development finance, since recipient choice for one form of donor financing rather than another underpins donors' strategies in response to competition. Beyond recipients' preferences for higher amounts of development finance with fewer conditions attached, which are discussed in the existing literature on competition with China (Hernandez 2017), I suggest recipients also have a preference for projects that are larger, and targeted at infrastructure. These preferences arise from the recipient government's political incentives.

Recipients' preferences with respect to the *type* of development interventions supported by outside finance stem from political imperatives to deliver tangible and visible development achievements. This underpins the preference for large projects, especially infrastructure-intensive projects in productive sectors, which are more readily communicated to voters than more diffuse development interventions, for example in governance or in the health sector. Governments in most developing countries derive their legitimacy from being able to deliver gains in development, which are easiest to demonstrate with highly visible interventions, such as investments in public infrastructure. For instance, in Ethiopia, a donor official noted that the government's investment in infrastructure "is also intended to strengthen the legitimacy of the party, which has been tarnished lately. These mega-projects, prestige projects...they're often about delivering a sense of development".¹³ Chinese financing is targeted precisely at such projects, providing funding for large public investments in infrastructure. As Swedlund (2017a: 404-405) and others point out, the process of disbursing Chinese finance, which tends to be rapid and streamlined, is also particularly suited to meeting political demands to demonstrate development through investment in infrastructure.

Financing for infrastructure is also attractive to recipient countries for the development model that it supports. China's own emphasis on a state-led infrastructure-intensive growth trajectory has inspired many governments in developing countries to pursue similar models. In Ethiopia, a Chinese official described the importance of this affinity between the two countries' approaches to development: "Ethiopia does not fully follow the Western model, they see the Chinese model as attractive. That's important for political dialogue, that they see the Chinese model can be good for Ethiopia".¹⁴ Even in developing countries with more liberal approaches to development than Ethiopia, the Chinese emphasis on state-led infrastructure investment has shaped expectations about development interventions. President Uhuru Kenyatta of Kenya recently praised the Chinese approach by saying, "The [Chinese] Belt and Road initiative gives our continent the opportunity to make a paradigm shift. Post-colonial Africa has been stuck in a rut" (Ilado 2017). Furthermore, as numerous studies on "infrastructure gaps" attest, financing for infrastructure is scarce, making China's provision of this type of development finance particularly attractive (Bank 2010; Serebrisky et al. 2015). Recipient preference for large, infrastructure-focused development finance thus also reflects, for some countries, an interest in a certain kind of development model, one that is premised on public investment in infrastructure.

Of course, large projects of the Chinese type can also be attractive to recipient country

¹³ Interview, donor representative. Addis Ababa, Ethiopia. July 26, 2017

¹⁴ Interview, Chinese representative. Addis Ababa, Ethiopia. August 7, 2017.

For a discussion of the embrace of China's development model in Ethiopia, see Fourie (2015)

governments for less developmentally-minded reasons, since they may provide more opportunities for corruption. Indeed, Brazys et al. (2017) find that Chinese development projects in Tanzania were associated with higher local experiences of corruption. In Kenya, there are frequent allegations of corruption in association with Chinese projects, as in a case where two Chinese state-owned telecommunications companies, Huawei and ZTE, bid for a contract to provide communications equipment to the police service in 2012, and the rival companies were alleged to be providing inducements to Kenyan political leaders (Mutegei 2013; Wahome 2012). Commenting on the incident, a Kenyan journalist said that “every politician wants a Chinese contractor on their side”.¹⁵ Thus, in some cases the attraction of Chinese-style development projects for recipient governments may be driven by narrower self-interested reasons, rather than political or developmental motivations. Traditional donors are likely less willing to adjust their development practice to meet these recipient preferences. Nevertheless, politically- and developmentally-motivated recipient preferences for Chinese finance mean that traditional donors could still face competitive pressure.

Political incentives thus lead recipients to prefer development finance that supports large-scale infrastructure investments, such as the type that Chinese official finance tends to support. However, developing country governments may not wish *all* of their development finance to be of this type. In fact, it is conceivable that the optimum outcome for recipient countries is a specialization among donors with respect to the development interventions they support, with traditional donors supporting projects in social sectors, including health, education, and governance, and infrastructure projects financed with Chinese support. Indeed, some government interview respondents suggested that traditional donors were better at projects in soft sectors and capacity building, and ought to continue to provide aid in these areas, while China expanded in hard sectors. For instance, a National Treasury official in Kenya commented that “the comparative advantage of traditional donors is in technical advice”.¹⁶ An official in Ghana similarly highlighted the value placed on traditional donors’ policy expertise: “If we completely go that way [commercial finance], we miss some of the good things from engagement with the traditional development partners, including the policy dialogue. We miss that from the emerging donors, we’re at risk of losing it”.¹⁷ In line with these claims about the benefits of different types of aid, Swedlund (2017a: 393) suggests “Chinese development assistance...performs a complementary role”.

This more varied set of recipient preferences with respect to the focus and type of development projects affects predictions about donors’ best responses to competition from China. Much of the existing work on donor competition has focused on those aspects of aid where recipients have a fairly linear utility function, preferring more aid to less (Barthel et al. 2014; Fuchs et al. 2015; Granath 2016), and preferring fewer conditions to more (Hernandez 2017). However, recipients may have more complicated preferences with respect to the *type* of development interventions that are supported. Larger and infrastructure-focused projects are preferred, but not all development interventions need be of this type. While the best response for donors competing on the quantity and conditionality of aid are fairly evident - provide more aid, attach fewer conditions - this is less straightforward when competition affects the *type* of development

¹⁵ Interview, Kenyan journalist. Nairobi, Kenya. January 25, 2017.

¹⁶ Interview, government official. Nairobi, Kenya. February 10, 2017.

¹⁷ Interview, government official. Accra, Ghana. May 2, 2017.

projects supported. There are two possible approaches that traditional donors can take when confronted with Chinese competition: they can emulate the Chinese approach to development finance and offer the same type of development projects to retain their attractiveness and access to the government, or they can choose to differentiate themselves by specializing in a complementary niche with their development projects, hoping that this is sufficient to retain their share of the recipient's external finance.

3.3 Incentives to emulate

Emulation may be the most direct strategy for donors to ensure they remain attractive to recipient governments and retain their access to recipient policymaking. As they observe Chinese financing winning over recipients with large infrastructure projects, donors may reason that the easiest way to retain “market share” is to offer the same kind of aid. Indeed, several recipient government representatives indicated in interviews they had observed or anticipated such a convergence, with traditional donors emulating Chinese development finance to remain competitive. In Kenya, a National Treasury official suggested, “Traditional donors will not just make space for the emerging partners...over time, the terms will change”.¹⁸ Donors also recognized this pressure for convergence. In Ethiopia, a donor representative suggested that those donors not yet providing infrastructure projects would have to do so to remain valuable to the government: “This was the Europeans’ mistake, that now China and Turkey come in here more and more. We didn’t think enough about financing for infrastructure, and that’s what governments need..If we continue this way, then we will become irrelevant”.¹⁹

Furthermore, there has been an increasing emphasis in development policy circles on the important of financing for infrastructure investment and economic transformation, in part driven by the example of China’s development experience (Lin and Wang 2017; UNCTAD 2014). This shift in policy thinking may provide a justification for increasing such financing in response to competitive pressure.

As a multilateral donor, the World Bank is particularly well placed among traditional donors to respond by emulating the Chinese approach and increasing the funding it dedicates to large, infrastructure-intensive projects. Bilateral donors may have less success with such a strategy. At the same time as China’s role as a provider of development finance has been increasing, many donor countries have reduced aid budgets and required their development agencies to specialize to achieve development impact with fewer resources. Though the World Bank is not immune to reductions in aid budgets and donor fatigue since it relies on contributions from shareholders for its concessional financing, it has a larger pool of resources from which to draw, and its IBRD borrowing window is able to rely on funds raised in international capital markets for its financing operations (Humphrey 2017). This gives it resources for the large projects that characterize Chinese development finance.

If donors respond to Chinese competition by *emulating* the Chinese approach to development finance, then I would expect to observe that recipient countries receiving more Chinese financing commitments receive more “Chinese-style” aid and finance from traditional donors in subsequent years: larger projects and more projects in infrastructure-intensive productive

¹⁸ Interview, government official. Nairobi, Kenya. February 13, 2017

¹⁹ Interview, donor official. Addis Ababa, Ethiopia. August 18, 2017.

sectors.

3.4 Incentives to differentiate

Yet, there are also reasons traditional donors may prefer to respond to the competitive threat of Chinese financing by instead specializing in sectors that are distinct from the Chinese approach. Emulating China’s approach to development projects may not always be possible with the resources available to traditional donors, who are operating with constrained aid budgets. As one donor representative in Nairobi noted, referring to the large railway project that China had recently funded in Kenya: “...unless you’re bringing in \$4 billion for the SGR [standard gauge railway], there’s no competing [with China]”.²⁰ Without the resources to offer the same type of projects, a better strategy may be to specialize.

Furthermore, for both the World Bank and the US, their comparative advantage relative to China is likely to lie in less resource-intensive policy advice and development projects targeted at softer sectors. In the case of the World Bank, it has seen its value for recipient countries in the technical expertise that it offers since at least the mid-1990s when James Wolfensohn defined the identity of the Bank as the “Knowledge Bank” (Kramarz and Momani 2013). In the case of the US, USAID has developed a strong profile in governance and social sector aid, specializing in health and education in many countries. This focus may mean that it is easier to respond to Chinese competition by developing a distinct niche that is sufficiently attractive to recipient countries, and where the US can outperform the Chinese competition.

If donors respond to Chinese competition by *differentiating* their development assistance, then I would expect to observe that recipient countries receiving greater commitments of Chinese development finance would subsequently receive smaller projects from traditional donors, as well as a smaller share of projects in infrastructure-intensive “hard” sectors, with more projects being allocated to “soft” sectors where these traditional donors might have a comparative advantage.

Given both of these conceivable sets of incentives, to emulate and to differentiate, it is not self-evident which strategy traditional donors will adopt in response to China’s growing role in development finance. The empirical tests below speak to that question.

4 Data and methods

My empirical analysis uses panel data on over 100 countries that have received Chinese official finance over the time period 2000 to 2014. My key explanatory variable is taken from the AidData Global Chinese Official Finance dataset and measures logged total financing commitments from China to a given country in a given year (Dreher et al. 2017a). Since other scholarship has highlighted the importance of disaggregating Chinese finance by the level of concessionality, I also present models in which I separate Chinese finance into aid-like concessional commitments and other, non-concessional financing commitments. In all models, I lag Chinese commitments by at least one year to account for the delay in the World Bank and the US responding to the announcement of Chinese financing commitments. Furthermore, to

²⁰ Interview, Donor Representative, African Development Bank. Nairobi, Kenya. February 2, 2017

account for the endogeneity of Chinese development finance to traditional donor behavior, I also present results using an instrument for Chinese finance.

I use two measures of traditional donor behavior for the dependent variables, both of which I calculate for the World Bank and the US. The first dependent variable measures the share of traditional donor development projects in a given year that are in “hard” sectors, such as transport, manufacturing, or telecommunications, as opposed to “softer” sectors, such as health or education. For the World Bank, I code the following as “hard” sectors: (1) energy and extractives, (2) industry, trade and services, and (3) transport. For the US, which uses a different classification scheme, I code these as “hard” sectors: (1) water and sanitation, (2) transport, (3) communications, (4) energy, (5) agriculture, (6) industry, mining and construction.²¹ I then calculate the proportion of annual development projects that are in these “hard” sectors and use this as the first dependent variable. The second dependent variable measures the number of large development projects (\$50-\$100 million and over \$100 million) committed by a donor in a given country in a given year. It captures whether traditional donors are willing to finance larger development projects in response to Chinese competition.²²

I use these two measures as a test of whether traditional donors are responding to competition through emulating or differentiating themselves vis-à-vis the Chinese approach to development. If traditional donors are choosing emulation as their strategy in response to competition, then I expect to observe a positive association between Chinese finance commitments and both dependent variables: greater commitments of Chinese finance should trigger a greater proportion of “hard” sector projects and a higher count of large projects from the World Bank or the US in subsequent years. By contrast, if traditional donors are seeking to differentiate themselves in response to competition, then I expect to observe a negative association between Chinese finance and the two dependent variables. In that case, the World Bank or the US would allocate a smaller share of a country’s development finance to “hard” sector projects, instead specializing in “soft” sectors, and extend fewer of the large projects.

I include a number of controls drawn from other models of development finance, and which are likely to be associated with the type of development projects that developing countries receive: GDP per capita (logged), external debt (as a percent of GDP), level of democracy, and recipient government effectiveness, with the expectation that more developed and more effective recipient governments should receive larger projects, and possibly more projects in “hard” sectors, while more indebted countries are likely to receive fewer of such projects.

²¹ The World Bank’s sector classification is different from that the OECD DAC classification used by the US, but I have made an effort to make these two as comparable as possible. The World Bank’s internal sector classification structure comprises eleven sectors, of which I code three as “hard” sectors, while the USAID dataset has 24 sector categories, of which I code six as “hard” sectors. Furthermore, World Bank projects may be associated with multiple sectors, while both the Chinese and USAID projects are only assigned to a single sector. To make these comparable, I code a World Bank project as being in a “hard” sector only if its one and only sector is one of the three “hard” sectors.

²² In the period under study, the average size of a World Bank project was \$76.96 million, with a standard deviation of \$167.81 million. The average number of World Bank projects of \$50-100 million per country per year is one project (mean: 0.50), and the average number of World Bank projects over \$100 million is also one (mean: 0.90), though there is a great deal of variation across the countries in the sample. For the US, projects are much smaller, and the average size of a US development project was \$1.7 million during the period under study, with a standard deviation of \$45.9 million. The average number of US projects of \$50-100 million per country per year is zero (mean: 0.27) and the average number of US projects over \$100 million per country per year is also zero (mean: 0.27)

I first estimate a series of panel OLS and Poisson regressions, with country- and year-fixed effects added in some models, and standard errors clustered by recipient country throughout. Subsequently, I adopt an instrumental variable approach to account for the fact that the Chinese decision to allocate finance to one recipient country rather than another is likely also related to the behavior of traditional donors. The instrument thus allows me to parse out the direction of the relationship: do higher Chinese financing commitments lead to changes in donor behavior, rather than the reverse?

The instrument, which has been applied to Chinese finance in other work by Dreher et al. (2017a), leverages the fact that Chinese overseas financing is in part driven by Chinese domestic factors that are exogenous to recipient countries, including steel production. Since steel is an input into the construction projects financed by Chinese development finance, greater steel production in China is an exogenous shock that leads to higher amounts of Chinese development finance in subsequent years. Steel production thus offers a time-varying instrument. To get at cross-country variation, lagged steel production is then multiplied with each country’s probability of receiving Chinese financing, which is calculated as the proportion of years in the sample (2000-2014) in which the country has received Chinese finance. With the inclusion of country- and year-fixed effects, this interacted instrument yields a plausibly exogenous instrument.

This set-up is similar to that of Nunn and Qian (2014) and Ahmed (2016), in which an exogenous time-varying instrument is interacted with an endogenous measure of cross-country variation to yield an instrument that varies over countries and over time, and which is rendered exogenous with the inclusion of country- and year-fixed effects. As Dreher et al. (2017a) explain, this approach follows a difference-in-difference intuition, evaluating “the differential effect of Chinese steel production on the amount of aid [finance commitments] to countries with a high compared to a low probability of receiving Chinese aid,” with the assumption that traditional donor behavior in countries with different probabilities of receiving Chinese financing commitments will not be affected differently by shifts in Chinese steel production, other than via the impact of Chinese finance.

Following the advice of Christian and Barrett (2017), I check this assumption by plotting Chinese steel production over time, and average Chinese official finance, traditional donor large project counts and traditional donor sectoral allocation by two groups: those above and below the median probability of receiving Chinese finance. These plots in the appendix (see Figure A1) suggest that the parallel trends assumption broadly holds for total Chinese financing commitments and for the share of projects in hard sectors, while it is less valid for the counts of large projects, where the trends for countries above and below the median probability of receiving Chinese finance do not move in parallel. I therefore only apply the instrumental variable approach to the analysis of the share of “hard” sector projects.

5 Results

Together, the results suggest that *emulation* is a more prominent strategy in response to competition from Chinese development finance than differentiation, and that the World Bank is more likely to adjust its development finance in response to Chinese commitments than the US. I find the World Bank increases the share of its projects in “hard” sectors in response to an increase

in Chinese financing commitments in a recipient country, which I interpret as evidence that the World Bank imitates “Chinese-style” development projects when China becomes a growing presence in a recipient country. There is some evidence the US also engages in emulation, increasing the number of large projects it extends to recipient countries in response to greater Chinese commitments.

Beginning with the relationship between Chinese financing commitments and World Bank behavior in Table 1, I find consistent evidence the World Bank is increasing the share of its development projects in “hard” sectors in response to commitments of Chinese finance, i.e. emulating a “Chinese-style” approach to development finance. There is also suggestive evidence the World Bank is imitating China in the case of large projects, increasing the number of large projects it funds in response to competition from Chinese finance, but these results do not hold in the most robust specifications, with country- and year-fixed effects included.

Columns 1-3 in Table 1 report a consistently positive and statistically significant relationship between Chinese financing commitments and the share of World Bank projects in infrastructure-intensive “hard” sectors, i.e. energy, industry, and transport. In the most robust specification, with country- and year-fixed effects included (column 3), a standard deviation increase in Chinese financing commitments (\$2.59 million (logged)) is associated with a 2% increase in the share of World Bank projects in “hard” sectors, all else equal. Though this appears to be a small effect, substantively, it could actually have meaningful consequences for recipient countries. Since such infrastructure-intensive projects tend to be expensive and politically salient, even a small increase in the share of this type of development projects could have a consequential effect for recipient governments.

The evidence on World Bank project size is less conclusive. Columns 4-6 in Table 1 report Poisson count models estimating the association between Chinese financing commitments and the number of World Bank projects costing more than \$100 million. In the first model, without country- and year-fixed effects, there is a positive and significant association between Chinese finance and the number of very large projects subsequently committed by the World Bank. Interpreted as an incidence rate ratio, an increase in Chinese financing commitments of one standard deviation (\$2.59 million (logged)) is associated with a 2.82 times greater rate of large projects, over \$100 million, *ceteris paribus*. However, as columns 5 and 6 show, these results do not hold with the inclusion of fixed effects, and suggest the first set of results are perhaps driven by selection effects. I estimate the same models for the count of World Bank projects valued at \$50-100 million (see Table A1), with the same finding: a positive and significant association that is not robust to the inclusion of fixed effects.

Across all models in Table 1, the controls behave largely as expected. A higher level of development (GDP per capita (log)) is associated with more projects in “hard” sectors, perhaps because immediate humanitarian and social needs are less pronounced in these countries, leading the World Bank to deliver projects targeted at productive sectors instead. Greater government effectiveness, as measured by the World Bank’s World Governance Indicators, is associated with a higher count of large projects, suggesting that governments seen as effective are more likely to be trusted with costly and complicated projects. Higher debt levels are associated with a smaller share of World Bank projects in “hard” sectors, and fewer large World Bank projects, as expected.

Table 1: OLS and Poisson Models: World Bank behavior and Chinese finance commitments

	WB: Hard sector share			WB: Count of large projects		
	(1)	(2)	(3)	(4)	(5)	(6)
Total Chinese finance commitments (log)	0.009*** (0.002)	0.008*** (0.003)	0.008*** (0.003)	0.083*** (0.025)	0.010 (0.009)	0.006 (0.009)
GDP per capita (log)	0.023** (0.009)	0.093*** (0.019)	0.051* (0.030)	-0.004 (0.160)	0.330*** (0.077)	-0.005 (0.197)
Government Effectiveness	0.006 (0.017)	0.002 (0.037)	0.008 (0.037)	0.953*** (0.316)	0.470** (0.204)	0.537*** (0.200)
Debt (% of GDP)	-0.025* (0.013)	-0.035** (0.017)	-0.033** (0.016)	-2.051*** (0.572)	0.120 (0.140)	0.070 (0.131)
Democracy	-0.001 (0.001)	-0.000 (0.003)	-0.001 (0.003)	-0.006 (0.027)	0.008 (0.018)	0.003 (0.018)
Observations	1,157	1,157	1,157	1,157	1,157	1,157
R-squared	0.035	0.201	0.226			
Country FE	NO	YES	YES	NO	YES	YES
Year FE	NO	NO	YES	NO	NO	YES
Nr. of countries	105	105	105	105	105	105

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

By contrast to the World Bank, there is no clear evidence the US is either emulating China or specializing in response to Chinese financing commitments. In Table 2, there is neither a statistically significant positive nor negative association between Chinese financing commitments and the two measures of US aid behavior. With respect to the share of US development projects in “hard” sectors such as transport or manufacturing, columns 1-3 in Table 2 show there is little evidence to believe the US is allocating a greater or lesser share of its aid projects to such sectors in response to competition from China. The negative association in column 1 is only statistically significant at the $p<0.10$ level, and falls away in columns 2 and 3 with the inclusion of fixed effects. There is also no statistically significant association between Chinese finance and the count of large US development projects (measured as projects over \$100 million). In Table A1 in the Appendix, I report results from models where the count of big projects is measured as the number of projects valued at \$50-100 million. Here there is some suggestion of a positive association between Chinese finance and the number of large and costly projects, but this relationship does not hold with the inclusion of fixed effects (see columns 5 and 6 in Table A1). Thus, if there is any evidence that the US is increasing the size of its development projects to emulate China, this is only with medium-sized projects, rather than the largest ones.

Table 2: OLS and Poisson Models: US aid behavior and Chinese finance commitments

	US: Hard sector share			US: Count of large projects		
	(1)	(2)	(3)	(4)	(5)	(6)
Total Chinese finance commitments (log)	-0.001* (0.001)	-0.001 (0.001)	-0.001 (0.001)	0.065 (0.066)	0.029 (0.029)	0.021 (0.028)
GDP per capita (log)	0.006* (0.003)	-0.008 (0.008)	-0.009 (0.014)	0.237 (0.226)	0.043 (0.235)	-0.250 (0.482)
Government effectiveness	0.032*** (0.007)	0.002 (0.011)	-0.003 (0.012)	0.472 (0.486)	-0.560 (0.665)	-0.723 (0.714)
Debt(% of GDP)	0.001 (0.004)	-0.012** (0.005)	-0.013** (0.006)	-1.058 (1.080)	-0.232 (0.294)	-0.037 (0.168)
Democracy	-0.000 (0.001)	0.000 (0.001)	-0.000 (0.001)	-0.031 (0.050)	-0.000 (0.025)	-0.015 (0.032)
Observations	1,404	1,404	1,404	1,404	1,404	1,404
R-squared	0.153	0.554	0.564			
Country FE	NO	YES	YES	NO	YES	YES
Year FE	NO	NO	YES	NO	NO	YES
Nr. of countries	104	104	104	104	104	104

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

In the next set of models, I probe the findings from Tables 1 and 2, and analyze how the concessionality of Chinese finance affects traditional donors' responses to Chinese competition. Previous work on Chinese development finance has emphasized the importance of distinguishing between different types of Chinese development finance, since not all Chinese finance is comparable to Western donors' official development assistance (ODA) (Brazys et al. 2017; Dreher et al. 2017a,b; Strange et al. 2017). As explained in Section 3, I anticipate that all Chinese finance could have a competitive effect on traditional donor behavior, since a recipient's choice to take up any Chinese finance, whether concessional or non-concessional, reduces the access traditional donors have to the recipient government. Indeed, the results in Table 1 suggest that for the World Bank, at least, overall Chinese finance is associated with a change in behavior. Nevertheless, since the *types* of development projects financed by concessional and non-concessional Chinese finance differ, with less concessional finance more likely to be channeled into larger and infrastructure-intensive projects, the pressure for traditional donors to emulate or differentiate the Chinese approach might differ.²³ To assess this, in Table 3 I use data from the AidData dataset on the terms of Chinese financing commitments to separate out concessional finance similar to ODA, "ODA-like" commitments, and less concessional finance, termed "other official flows" or "OOF-like" commitments.

Table 3 shows that the association between Chinese finance and traditional donor behavior by concessionality differs by donors. While the World Bank appears more likely to change their behavior in response to non-concessional OOF-like Chinese finance, the US seemingly adjusts its behavior and imitates a "Chinese-style" approach to development in response to

²³ OOF-like Chinese projects are more likely to be in hard sectors: of the 511 OOF-like projects for which there is sector data, 57.34% are in "hard sectors", while only 23.56% of the 1,982 ODA-like Chinese projects in the AidData dataset are in sectors I have classified as "hard" sectors.

concessional ODA-like commitments. In the case of the World Bank, the finding from Table 1 that the Bank increases the share of development projects in infrastructure-intensive sectors in response to Chinese financing commitments appears to be driven OOF-like commitments, rather than concessional ODA-like commitments. In column 1 in Table 3, OOF-like commitments are statistically significantly associated with the World Bank allocating a greater share of projects to “hard” sectors, while there is no such relationship for concessional ODA-like commitments. This suggests that for the World Bank, the competitive pressure to converge on “Chinese-style” development practices comes from OOF-like commitments. While the World Bank may be shifting its projects into “hard” sectors and emulating China in response to competition, there is no evidence the World Bank is either emulating or differentiating itself from China with respect to the size of projects, see columns 4-5 in Table 3.

In the case of US aid, by contrast, there is evidence that the US is more responsive to ODA-like Chinese commitments than non-concessional OOF-like commitments, increasing the number of large development projects when recipient countries receive more ODA-like Chinese finance. Thus, the US appears more inclined to mimic the Chinese approach to development by offering large projects when Chinese commitments are ODA-like and concessional. While there is no statistically significant association between Chinese commitments and the sectoral allocation of US aid (columns 3 and 4 in Table 3), there is a statistically significant positive association between Chinese ODA-like commitments and the subsequent number of large development projects committed by the US. A difference of one standard deviation in Chinese ODA-like commitments (\$1.96 million (logged)) is associated with a 2.11 times greater rate of US aid projects over \$100 million, all else constant.

While emulation responses on the part of the World Bank are more likely to be triggered by OOF-like finance, US aid appears emulate Chinese development approaches in response to concessional ODA-like commitments. The different results for the World Bank and the US perhaps reflect their different abilities to emulate Chinese finance, and different comparative advantages. The World Bank’s lending arm, the International Bank for Reconstruction and Development, allows it to provide loans that are more similar to Chinese development loans, while the US’ specialization in grant aid might explain why it finds it more strategic to respond to Chinese ODA-like commitments that could be a potential substitute for US aid in recipient countries.

Table 3: OLS and Poisson models: World Bank and US behavior by concessionality of Chinese finance

	WB: Hard sector share		US: Hard sector share		WB: Large projects		US: Large projects	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Chinese OOF-like commitments (log)	0.008** (0.003)		0.000 (0.001)		0.002 (0.010)		0.013 (0.022)	
Chinese ODA-like commitments (log)		0.005 (0.004)		-0.001 (0.001)		-0.007 (0.019)		0.072** (0.034)
GDP per capita (log)	0.063** (0.029)	0.069** (0.029)	-0.012 (0.014)	-0.012 (0.014)	0.285 (0.256)	0.284 (0.257)	-0.487 (0.545)	-0.504 (0.544)
Government effectiveness	0.007 (0.039)	0.013 (0.038)	-0.005 (0.013)	-0.005 (0.013)	0.544** (0.253)	0.542** (0.249)	-0.863 (0.704)	-0.809 (0.688)
Debt (as % of GDP)	-0.037** (0.015)	-0.038*** (0.014)	-0.014** (0.007)	-0.014** (0.007)	0.342* (0.198)	0.340* (0.203)	-0.014 (0.182)	0.008 (0.196)
Democracy	-0.000 (0.003)	-0.001 (0.003)	-0.000 (0.001)	-0.000 (0.001)	-0.005 (0.022)	-0.005 (0.022)	-0.040 (0.036)	-0.049 (0.036)
Observations	1,073	1,073	1,307	1,307	1,073	1,073	1,307	1,307
R-squared	0.237	0.233	0.562	0.563				
Country FE	YES	YES	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES	YES	YES
Nr. of countries	105	105	104	104	105	105	104	104

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Finally, I use an instrumental variable approach to verify the most consistent finding from the previous models, namely the positive association between Chinese financing commitments and the proportion of projects the World Bank subsequently allocated to “hard” sectors, which I see as evidence of the World Bank emulating the Chinese approach to development finance in response to competition. As explained in Section 4 above, the instrumental variable approach uses Chinese steel production and the probability of receiving Chinese finance to as an instrument that varies over countries and years.

Table 4 reports the results of a series of two stage-least-squares (2SLS) models of the proportion of World Bank projects allocated to “hard” sectors, with Chinese finance commitments broken down by concessionality. The interacted instrument is a strong predictor of overall Chinese finance commitments and OOF-like commitments (columns 1 and 2, F statistic greater than 30), but not for Chinese ODA-like commitments (column 3, F statistic: 4.24). This is not surprising, since Chinese steel is more likely to be incorporated in the infrastructure projects financed by OOF-like finance than in the ODA-like grant-funded projects. For each of the three specifications, a Wooldridge test suggests that we can reject the null hypothesis of exogenous regressors, and an instrumental variable approach is appropriate (For total Chinese finance, column 1, p=0.049; for Chinese OOF-like commitments, column 2, p=0.057; for Chinese ODA-like commitments, column 3, p=0.036)

There is a positive and statistically significant relationship across the models between Chinese financing commitments and the World Bank’s subsequent share of projects in “hard” sectors.²⁴ Given the instrumental approach, we can be more confident that this relationship is in the direction that I suggest, namely that the World Bank is responding to Chinese commitments

²⁴ These results hold when the control variables are dropped on robustness checks, since there is a risk that the control variables are acting as “bad” post-treatment controls (Angrist and Pischke 2008).

through emulation, rather than in the opposite direction, in which Chinese finance is allocated to countries that receive more “hard” sector projects from the World Bank. In the case of the World Bank, there is consistent evidence that Chinese financing leads to a shift in the *type* of development projects supported, with an increase in the projects in infrastructure-intensive or productive “hard” sectors in response to Chinese development finance.

The coefficients of the 2SLS estimates in Table 4 are larger than the coefficients in the OLS estimates in Tables 1 and 3. There is no strong expectation about the direction of bias in the OLS estimates: China might be providing financing precisely to those countries not receiving many “hard sector” projects from the World Bank, filling a gap, or it might be choosing the same countries to extend financing to as the World Bank gives “hard sector” projects to. The larger effects in the 2SLS models in Table 4 suggest that the bias in the OLS models was negative, i.e. that China is allocating finance to those countries receiving fewer “hard sector” projects from the World Bank. This finding is in line with other work on Chinese finance, such as Dreher et al. (2017a: 20) who also find larger 2SLS effects than OLS effects of Chinese finance on economic growth, which they explain with reference to the fact that China allocates more finance to countries experiencing low growth.

Table 4: Instrumental variable 2SLS estimation: World Bank hard sector shares

	WB: Hard sector shares		
	(1)	(2)	(3)
Total Chinese finance commitments (log)	0.075** (0.038)		
Chinese OOF-like commitments (log)		0.052** (0.024)	
Chinese ODA-like commitments (log)			0.097** (0.049)
GDP per capita (log)	0.048 (0.040)	0.035 (0.038)	0.087** (0.039)
Government Effectiveness	0.012 (0.046)	-0.017 (0.046)	0.049 (0.047)
Debt (as % of GDP)	-0.038** (0.018)	-0.033* (0.020)	-0.046*** (0.015)
Democracy	-0.002 (0.004)	0.001 (0.003)	-0.004 (0.005)
Observations	1,073	1,073	1,073
Country FE	YES	YES	YES
Year FE	YES	YES	YES
First-stage F statistic	32.40	48.79	4.24

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Together, these results suggest that the World Bank, in particular, is responding to the growing presence of China as a source of development finance at the country level. The results indicate that where there is a response to Chinese finance, it is one of emulation, rather than differentiation: the World Bank is moving more of its aid projects into the types of sectors where

China is also active, rather than specializing, and there is some evidence the US is increasing the number of large project it commits in countries receiving large amounts of Chinese finance, rather than reducing large projects to focus on smaller development interventions distinct from the Chinese approach.

6 Conclusion

The arrival of China as a substantial provider of development finance has shaken up the practices of international development. It provides recipient countries with an additional source of development finance, and emphasizes different aspects of development, stressing investments in productive sectors and infrastructure.

I argue that China's growing role as a provider of development finance introduces competitive pressures on traditional donors, who risk losing access to recipient governments as China becomes an increasingly dominant player. I suggest that traditional donors can respond to competition from China either by imitating China's approach to development, thereby hoping to remain attractive to recipient governments, or by specializing in a complementary niche that highlights their relevance to recipient governments.

The analysis of the World Bank and US's responses to Chinese finance commitments suggest that emulation is the more common strategy among these donors. When countries receive higher commitments of Chinese finance, the World Bank increases the share of its projects in precisely those infrastructure-intensive "hard sectors" that make up the bulk of Chinese finance. While the World Bank increasingly emulates Chinese development practices, the US is less responsive to Chinese aid commitments, though there is some evidence that the US increases the number of large development projects in response to China's ODA-like commitments.

These findings raise a number of questions for future research. It is striking that the World Bank appears more likely to change its development practice in response to Chinese competition than the US, given that bilateral development aid is often considered more subject to political influence than multilateral aid (Milner 2006; Milner and Tingley 2012). Is it that the World Bank is more threatened by competition from Chinese finance because Chinese-financed infrastructure projects are a more ready substitute for the World Bank's infrastructure projects than bilateral aid in governance or social sectors? Or is it that the World Bank's recent emphasis on economic transformation and industrialization in developing countries has allowed it to emulate the Chinese approach in a way that is not possible for the US, where domestic political expectations might require a pro-poor and social sector focus for development aid? Furthermore, is the World Bank representative of multilateral development banks? Are the African Development Bank and the Inter-American Development Bank, for example, also responding to Chinese competition through emulation? How is China exercising its voting rights within the World Bank with respect to this competition?

The observed pattern of traditional donor emulation in response to country-level Chinese presence is compatible with my theory of donor competition, building on previous work on donor competition. However, it is also conceivable that the changes in traditional donor behavior after China has committed finance to a recipient country is driven by non-competitive mechanisms:

China's engagement could signal that this recipient country is suitable for a particular type of development intervention, or could create positive externalities that make it easier to carry out infrastructure-intensive projects. Separating out these causal mechanisms may be best done with qualitative data, and in future work I plan to use interview evidence to analyze the motivations behind donors' responses to China's growing role.

This paper finds that traditional donors change their behavior in response to China's growing presence in developing countries. These findings have considerable implications for the relationship between donors and recipients, as well as for understanding China's impact on the governance and practices of international development. The increasing magnitude of Chinese development finance means that different types of development projects are being financed, providing an alternative to the approach of traditional donors. Furthermore, by acting as a competitor, China may alter the development practices of traditional donors in recipient countries in a "bottom-up" fashion.

As China continues to assert its growing strength in world politics, currently dominant actors and institutions must consider their responses. In some cases, emulation and convergence may be the best strategies to respond to competition, but in others, specialization and complementarity might leave all actors better off. Evidence from development finance suggests emulation is an emerging strategy adopted by existing institutions. Future work should consider the patterns of response in other issue-areas where emerging economies' influence is growing, such as environmental and social safeguards of construction projects, export finance, or investment.

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Table A1: Poisson models of the count of World Bank and US projects of \$50-100 million

	World Bank: Big projects			World Bank: Big projects		
	(1)	(2)	(3)	(4)	(5)	(6)
Chinese finance commitments (log)	0.075*** (0.020)	0.010 (0.009)	0.006 (0.009)	0.121*** (0.046)	0.045 (0.029)	0.033 (0.025)
GDP per capita (log)	-0.104 (0.107)	0.330*** (0.077)	-0.005 (0.197)	0.015 (0.192)	0.271 (0.270)	-0.118 (0.529)
Government Effectiveness	0.836*** (0.246)	0.470** (0.204)	0.537*** (0.200)	0.174 (0.351)	0.138 (0.590)	0.115 (0.590)
Debt (as % of GDP)	-1.574*** (0.383)	0.120 (0.140)	0.070 (0.131)	-0.998* (0.522)	-0.186 (0.172)	-0.169 (0.162)
Democracy	-0.001 (0.021)	0.008 (0.018)	0.003 (0.018)	0.022 (0.031)	0.047 (0.034)	0.040 (0.042)
Observations	1,157	1,157	1,157	1,404	1,404	1,404
Country FE	NO	YES	YES	NO	YES	YES
Year FE	NO	NO	YES	NO	NO	YES
Nr. of countries	105	105	105	104	104	104

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Figure A1: Parallel trends plots for the instrument, using an interaction of Chinese steel production and countries' probability of receiving Chinese finance

